frozen · gradient-trained · closed form or fixed (no gradient step) · *lever* what the method trains

H1 Encoder adaptation

five established methods and one input-side variant; all read the last layer with a gradient-trained head

LoRA



Trains: low-rank branch B·A in parallel to W_0 (q, v projections), plus the head.

DoRA



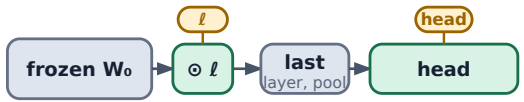
Trains: the same branch on the direction of W_0, plus a per-channel magnitude m, plus the head.

AdaLoRA



Trains: a low-rank branch in singular-value form whose rank budget moves across sites during training, plus the head.

IA³



Trains: a per-channel gate ℓ on the output of each projection (no rank), plus the head.

HiRA



Trains: a low-rank product multiplied element-wise by W_0, giving a high-rank update at LoRA's budget, plus the head.

Radial-LoRA — input-side variant

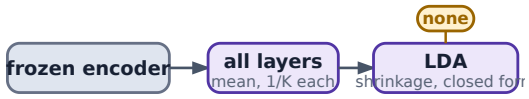


Trains: LoRA's branch on a norm-tempered input $g\lambda(u)\cdot u$; λ is fixed by rank (0.35 at r=8), not trained. Same budget as LoRA.

H2 The readout

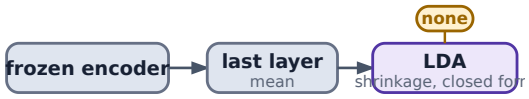
do not adapt; read the frozen encoder in closed form (proposed)

DFA



Trains: nothing. Equal-weight (1/K) fusion of all K = L+1 depths, standardisation, Ledoit–Wolf LDA fitted once.

DFA-last — depth control



Trains: nothing. Same head on the last layer only; isolates what depth fusion adds.

Controls

probe depth of the read, estimator and encoder adaptation

LoRA, all-layer read — `lora-ln-dw`



Trains: as LoRA, but the head reads the fixed uniform average of all layers. `lora-recipe` adds test-time checkpoint averaging.

Learned layer weights — `wls-readout`



Trains: the layer weights and the head by gradient on the frozen encoder (SUPERB-style); stage 1.